Information and learning in pseudo-visibility environments

Private annotations, shared randomness, and a tabular convergence theorem

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Abstract

A pseudo-visible character can be observed, but an agent is penalized for acting differently from its counterfactual action with that character erased. Alexander proposes exposing that action to a learning agent through a self-reflection wrapper. We separate three questions: what information an annotation reveals, which policies can be represented, and what a specified learner converges to. For an exogenous private query we give exact optimal values and a criterion for strict improvement. Shared random seeds need not create this information advantage. In a finite one-step model with greedy queries, sample-average tabular learning converges to the same policy with or without annotations under explicit sampling and uniqueness assumptions. That policy can be strictly suboptimal for the joint policy-dependent objective. Retained simulations illustrate the distinction between training reward, final greedy value, and representation effects.

1. The question behind the wrapper

A game character may see a player while being rewarded for behaving as though the player were invisible. Alexander calls this *pseudo-visibility* [A]. The environment compares the character's actual action with the action its policy would have chosen from an observation with the player erased. A discrepancy incurs a penalty.

This creates a learning problem. A learner might discover that a certain action is penalized without recognizing that the relevant feature is agreement with its own counterfactual behavior. The SelfRefl wrapper in [A, Definition 10] supplies that feature: it annotates an observation with the policy's action on the erased observation, or a dummy marker when no annotation is needed. The source asks whether this improves performance, while recognizing that an agent may already perform the relevant reflection.

There are several possible meanings of improvement. An annotation might reveal information otherwise unavailable at action time. It might make a useful interaction expressible in a restricted feature class. Or it might change the speed or limit of a learning algorithm. These require different statements.

We first hold a private query's law fixed and solve its one-step decision problem. We then study a different setting in which the query is the learner's own greedy erased action and changes during training. The latter setting explains how equal learning limits can coexist with a substantial gap to the joint optimum.

2. Policies, annotations, and randomization

Write o for an observation and $e(o)$ for its erased version. Suppressing the dummy marker on ordinary observations, the wrapper has the form

$$h(o) = g(o, c(o)), \qquad c(o) = g(e(o), -1).$$

The underlying agent is called on an annotated erased observation to compute the counterfactual action, and is then called on the annotated actual observation. The source's Proposition 12 gives the corresponding environment-side presentation.

Proposition 1. Unrestricted deterministic representation

For every deterministic policy π, an underlying policy g can be chosen so that its wrapped policy h equals π.

Proof. Define $g(o, c) = \pi(o)$ for every annotation c, including the dummy marker. The actual-action call then returns $\pi(o)$, regardless of the counterfactual annotation. □

This elementary observation concerns the unrestricted function class. It supplies no equality of learning times or of restricted linear representations.

Randomness requires an additional distinction. The source suggests holding a policy's pseudo-random generator state fixed between training updates [A, discussion after Lemma 4]. Repeated calls then have a reproducible coupling. This does not imply independence between calls on different observations.

For example, retain a fair bit Z for the whole step and return Z at every observation. The erased and actual actions both have uniform marginal distributions, yet they agree with probability one. A lower bound on each marginal action probability therefore gives no positive mismatch bound by itself.

In contrast, a query drawn using private randomness independent of the plain agent's action may convey useful information when revealed. The next theorem states that decision problem explicitly. It is not a consequence of the frozen-generator convention alone.

3. Exact value of a private annotation

Fix a finite action set $\mathcal{A}$, a penalty $q \geq 0$, and a query C with distribution p_c. The plain decision maker does not observe C; the annotated one does. For each action a, base reward has conditional mean $R(a)$ independent of C. The one-step objective is

$$R(a) - q\mathbf{1}[a \neq C].$$

The query law is fixed while comparing policies. A plain randomized action uses no randomness correlated with C. An annotated policy may condition its action on the revealed query.

Theorem 2. Optimal one-step values

Under these assumptions,

$$V_{\text{plain}} = \max_a \{R(a) - q(1 - p_a)\},$$

$$V_{\text{ann}} = \sum_c p_c \max_a \{R(a) - q\mathbf{1}[a \neq c]\}, \qquad V_{\text{ann}} \geq V_{\text{plain}}.$$

Proof. A fixed plain action a mismatches with probability $1 - p_a$. Randomizing among actions forms a convex combination of their values, so a maximizing pure action suffices. After observing $C = c$, the annotated policy maximizes the corresponding conditional objective. Averaging those optima gives the second formula. An annotated policy can use any fixed plain action, proving the inequality. □

There is an exact equality criterion. The values are equal precisely when some single action maximizes $R(a) - q\mathbf{1}[a \neq c]$ for every c of positive probability. Indeed, choose a maximizing plain action. Its conditional shortfall from the annotated optimum is nonnegative in every case; its mean vanishes exactly when all positive-probability shortfalls vanish. Thus additional information has positive value when no action is optimal in all of the possible conditional problems.

Corollary 3. Equal base rewards

If $R(a) = R_*$ for all a, the gain is exactly

$$V_{\text{ann}} - V_{\text{plain}} = q(1 - \max_a p_a).$$

If there are m actions and every $p_a \geq \delta$, it is at least $q(m-1)\delta$.

Proof. Copying C achieves R_*, which is optimal. The best plain action has the largest p_a, hence the smallest mismatch probability. Finally $\max_a p_a \leq 1 - (m-1)\delta$. □

The gain is strict when $q > 0$ and the query is not concentrated on one action. Equal base rewards matter. For two actions, uniform C, $q = 1/4$, and $R(0) = 1, R(1) = 0$, the conditional values are

Revealed query	Choose 0	Choose 1
$C = 0$	1	$-1/4$
$C = 1$	$3/4$	0

Action 0 is optimal in both rows. Both informed and uninformed optima are $7/8$; always copying the query gives only $1/2$. Avoiding a mismatch need not compensate for lost base reward.

The theorem optimizes the visible action while holding the query law fixed. If changing the whole policy also changes that law, the objective is different. The next sections make that dependence part of the model.

4. A finite model with policy-dependent queries

An episode has a context x from a finite set and a flag $y \in \{0, 1\}$, indicating an erased or pseudo-visible observation. Write $R(x, y, a)$ for base-reward means. The query at a visible observation is the current greedy action at the corresponding erased observation.

The plain learner sees (x, y). The annotated learner sees $(x, 0, -1)$ or $(x, 1, c)$, where c is that query. Each keeps a separate tabular estimate for every observed state/action pair. Updates are sample averages of received rewards. There is no multi-step bootstrapping.

For a deterministic final policy, write $a_0(x)$ for its erased action and $a_1(x)$ for its visible action. If the visible flag has probability p, its value at context x is

$$(1-p)R(x,0,a_0(x)) + p\{R(x,1,a_1(x)) - q\mathbf{1}[a_1(x) \neq a_0(x)]\}.$$

The erased choice affects the visible penalty. An update based only on rewards received at erased observations does not directly charge that later effect to the erased action.

Theorem 4. Convergence of greedy-query tabular learning

Assume finitely many contexts and actions. Context/flag pairs are sampled independently across episodes from a fixed full-support law. Return noise is conditionally mean zero given the past, current state and chosen action, with uniformly bounded conditional second moments. Each action is visited infinitely often at each state that continues to occur. Use sample-average updates.

For each context assume unique maximizers

$$a_0^*(x) = \operatorname*{argmax}_a R(x,0,a), \qquad a_1^*(x) = \operatorname*{argmax}_a \{R(x,1,a) - q\mathbf{1}[a \neq a_0^*(x)]\}.$$

With greedy erased queries, the final greedy policies of the plain and annotated learners converge almost surely to (a_0^*, a_1^*).

Proof. Erased observations incur no visibility penalty. For each erased state/action pair, the sample-average martingale law of large numbers gives convergence to $R(x,0,a)$. Uniform conditional second-moment control suffices because $\sum_k k^{-2} < \infty$. Infinite visits and uniqueness make the greedy erased action eventually equal $a_0^*(x)$. There are finitely many contexts, so this stabilization occurs simultaneously after an almost surely finite time.

Thereafter every visible query at x equals $a_0^*(x)$. For the plain learner, visible reward means are now $R(x,1,a) - q\mathbf{1}[a \neq a_0^*(x)]$. For the annotated learner, the only continuing visible state is $(x,1,a_0^*(x))$, with the same means. Infinite action visits give convergence of both tables there. The finitely many earlier samples vanish in the averages. The unique visible maximizer therefore eventually equals $a_1^*(x)$ for both learners. Annotation states that cease occurring have no effect on the eventual policy. □

The conclusion concerns greedy policies used for evaluation. With persistent exploration, training actions need not converge to those policies, and training reward need not equal their value. Constant step sizes, tied maxima, shared features, sampled queries and general multi-step environments are outside this theorem.

5. Equal limits can be jointly suboptimal

Take one context, $p = 1/2, q = 2$, and

$$R_0 = (1, 0.9), \qquad R_1 = (0, 1).$$

The unique greedy erased action is 0. Given query 0, visible action 0 has value 0 and action 1 has value -1. Theorem 4 therefore selects $(0,0)$, of total value $1/2$.

The full deterministic objective has only four possibilities:

Erased action	Visible action	Joint value
0	0	0.50
0	1	0
1	0	−0.55
1	1	0.95

The joint optimum sacrifices 0.1 in erased-state reward to make the preferred visible action agree with the query. Its value is 0.95, a gap of 0.45 from the limiting greedy policy. An annotation exposes the query, but it does not by itself modify the erased state's credit assignment.

This supplies a restricted answer to the source's informal performance question: equality of these learning limits is compatible with substantial suboptimality. It does not say that every learning rule has the same limit or that annotations cannot improve finite-training performance.

6. What the retained experiments measure

The included program uses 30 seeds, numbered 0–29, with $\varepsilon = 0.1$ exploration. It reports mean training reward and exact value of each final greedy policy separately. The coupled example uses 4,000 episodes and Gaussian base-reward noise with standard deviation 0.3.

Query mode	Plain training mean	Annotated training mean	Both final values
Greedy	0.4679	0.4716	0.5000
Sampled	0.4220	0.4936	0.5000

In sampled mode, the code uses a separate query random stream and the annotated arm reuses its annotation as the environment's query answer. This implements a private-query protocol. It does not implement every coupling compatible with the source's frozen-generator advice. The sampled row is outside Theorem 4.

A second scenario has 40 contexts, four actions, uniform base means fixed using seed 56, a visible-state bonus of 0.4 to one action per context, $p = 0.4, q = 0.6$, and Gaussian noise standard deviation 0.5. The following entries are mean final greedy values from the saved record.

Episodes	Learner	Plain	Annotated
2,000	Tabular, sample average	0.6309	0.6521
20,000	Tabular, sample average	0.8010	0.8164
2,000	Linear, step 0.05	0.6644	0.7486
20,000	Linear, step 0.05	0.8306	0.8379

The enumerated joint optimum is approximately 0.8565; the greedy erased/visible best-response value is 0.8436. The latter describes a decision rule, not a proven convergence result for every learner in the table.

The linear learner uses context/action and flag/action features. The annotated version adds the shared indicator $\mathbf{1}[a = c]$. This changes the available feature list, so its finite-horizon gains cannot

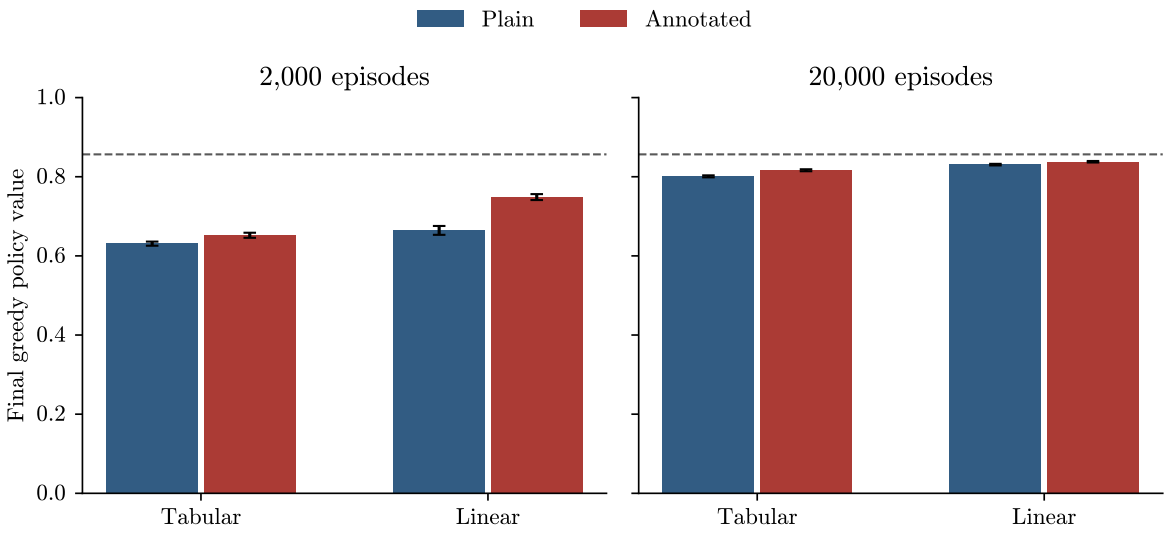


Figure 1: Final greedy values at two retained training horizons. Error bars are the record's standard errors across 30 seeds, not confidence intervals or a test of the paired difference. The dashed line is the enumerated joint optimum.

be interpreted solely as an effect on an otherwise identical representation. The saved record also includes constant-step tabular and sampled-query variants. No new training runs were performed for this paper.

7. Remaining questions

A learning-rate theorem needs an explicit learner, feature class, context distribution, accuracy target and confidence level. An assertion of constant versus context-linear sample complexity cannot be assessed without those inputs. The simulations provide finite-horizon comparisons, not such a theorem.

Theorems 2 and 4 also answer different questions. The first values private information under a fixed query distribution. The second follows a specified learning process whose greedy query eventually stabilizes. Neither establishes a universal advantage for the SelfRefl wrapper over all agents, and neither is a theorem about general policy-dependent MDPs.

The principal lesson for a proposed formal performance claim is to specify the query/action coupling, accessible information, policy class, training updates and evaluation objective together. The proofs and examples above make those choices inspectable.

References

[**A**] Samuel Allen Alexander. *Pseudo-visibility: a game mechanic involving willful ignorance*. Proceedings of FLAIRS-35 (2022). DOI: 10.32473/flairs.v35i.130652. Definitions 2 and 9–10, discussion after Lemma 4, Proposition 12, and the concluding better-performance discussion.